

Pythagoras, mensuration and 3D - Set 1 - Solutions

Specification M5.7, M5.11-M5.13, M5.15-M5.17.

Answer key

Question	Answer	Topic
1	B	Pythagoras, mensuration and 3D
2	C	Pythagoras, mensuration and 3D
3	E	Pythagoras, mensuration and 3D
4	F	Pythagoras, mensuration and 3D
5	B	Pythagoras, mensuration and 3D
6	E	Pythagoras, mensuration and 3D
7	D	Pythagoras, mensuration and 3D
8	F	Pythagoras, mensuration and 3D
9	E	Pythagoras, mensuration and 3D
10	B	Pythagoras, mensuration and 3D
11	F	Pythagoras, mensuration and 3D
12	F	Pythagoras, mensuration and 3D

Worked solutions

Question 1 MEN-01 · Pythagoras, mensuration and 3D · spec M5.7

A cuboid measures 3 cm by 4 cm by 12 cm. Find the length of its longest diagonal.

Answer: B

$$\sqrt{9 + 16 + 144} = \sqrt{169} = 13.$$

Question 2 MEN-03 · Pythagoras, mensuration and 3D · spec M5.16

A trapezium has parallel sides of length 8 cm and 14 cm and area 77 cm^2 . Find the perpendicular distance between the parallel sides.

Answer: C

$$\frac{1}{2}(8 + 14)h = 11h = 77 \Rightarrow h = 7.$$

Question 3 MEN-04 · Pythagoras, mensuration and 3D · spec M5.16

A cone has base radius 6 cm and slant height 10 cm. Find its volume. (Volume of a cone $= \frac{1}{3}\pi r^2 h$.)

Answer: E

$$\text{Height} = \sqrt{100 - 36} = 8. \text{ Volume} = \frac{1}{3}\pi \cdot 36 \cdot 8 = 96\pi.$$

Question 4 MEN-05 · Pythagoras, mensuration and 3D · spec M5.17

A sphere and a cube have the same volume. The cube has side length a . Find the radius of the sphere. (Volume of a sphere $= \frac{4}{3}\pi r^3$.)

Answer: F

$$\frac{4}{3}\pi r^3 = a^3 \Rightarrow r^3 = \frac{3a^3}{4\pi} \Rightarrow r = a\sqrt[3]{\frac{3}{4\pi}}.$$

Question 5 MEN-06 · Pythagoras, mensuration and 3D · spec M5.16

A square of side 10 has a quarter-circle of radius 10 drawn inside it, centred at one corner. Find the area of the square not covered by the quarter-circle.

Answer: B

$$100 - \frac{1}{4}\pi \cdot 100 = 100 - 25\pi.$$

Question 6 MEN-07 · Pythagoras, mensuration and 3D · spec M5.7

A ladder of length 6.5 m leans against a vertical wall with its foot 2.5 m from the wall on horizontal ground. How far up the wall does it reach?

Answer: E

$$\sqrt{6.5^2 - 2.5^2} = \sqrt{42.25 - 6.25} = \sqrt{36} = 6.$$

Question 7 MEN-08 · Pythagoras, mensuration and 3D · spec M5.18

In a right-angled triangle the hypotenuse is 10 and one angle is 30° . Find the area of the triangle.

Answer: D

$$\text{Sides } 10\sin 30^\circ = 5 \text{ and } 10\cos 30^\circ = 5\sqrt{3}. \text{ Area} = \frac{1}{2} \cdot 5 \cdot 5\sqrt{3} = \frac{25\sqrt{3}}{2}.$$

Question 8 MEN-09 · Pythagoras, mensuration and 3D · spec M5.16

The volume of a cube is numerically equal to its surface area. Find the length of its side.

Answer: F

$$a^3 = 6a^2 \Rightarrow a = 6.$$

Question 9 MEN-11 · Pythagoras, mensuration and 3D · spec M5.16

Which of the following statements is/are true?

- I Doubling the radius of a circle doubles its circumference.
- II Doubling the radius of a sphere multiplies its volume by 8.
- III Doubling the radius of a cylinder while halving its height leaves its volume unchanged.

Answer: E

$$C = 2\pi r \text{ is linear in } r \text{ (I). } V \propto r^3 \text{ (II). } \pi(2r)^2 \cdot \frac{h}{2} = 2\pi r^2 h: \text{ doubled, not unchanged (III false).}$$

Question 10 MEN-12 · Pythagoras, mensuration and 3D · spec M5.16

A running track consists of two straights of length 100 m and two semicircular ends of radius r . The total length of the track is 400 m. Find r .

Answer: B

$$200 + 2\pi r = 400 \Rightarrow r = \frac{100}{\pi}.$$

Question 11 MEN-14 · Pythagoras, mensuration and 3D · spec M5.16

A cylindrical tin of radius 4 cm and height 10 cm is full of water. The water is poured into an empty cylindrical jug of radius 5 cm. How deep is the water in the jug?

Answer: F

$$\pi \cdot 16 \cdot 10 = \pi \cdot 25 \cdot h \Rightarrow h = \frac{160}{25} = 6.4.$$

Question 12 MEN-15 · Pythagoras, mensuration and 3D · spec M5.17

A solid consists of a hemisphere of radius 3 on top of a cylinder of radius 3 and height 4. Find its volume. (Sphere volume $\frac{4}{3}\pi r^3$.)

Answer: F

$$\text{Cylinder } \pi \cdot 9 \cdot 4 = 36\pi; \text{ hemisphere } \frac{1}{2} \cdot \frac{4}{3}\pi \cdot 27 = 18\pi. \text{ Total } 54\pi.$$